MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC- CS502/PCC-CS502/PCCCS502/PCCCS503/PCCCSD503 Operating Systems

UPID : 005507

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) Every address generated by the CPU is divided into two parts. They are ____ and ____.

(ii) Logical memory is broken into blocks of the same size called ________.

(iii) The multi-user Operating System is based on the concept of _______.

(iv) Context switching is part of ________.

(v) The segment of code in which the process may change common variables, update tables, write into files is known as ____________.

(vi) CPU fetches the instruction from memory according to the value of ______.

(vii) In _________ method each file occupy a set of contiguous block on the disk.

(viii) Each entry in a segment table has a ___________.

(ix) The switching of the CPU from one process or thread to another is called ___________.

(x) A binary semaphore is a semaphore with integer values ____________.

(xi) To ensure difficulties do not arise in the readers – writers problem _______ are given exclusive access to the shared object.

(xii) When device A has a cable that plugs into device B, and device B has a cable that plugs into device C and device C plugs into a port on the computer, this arrangement is called a _________.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is a process? With the help of a diagram, explain the different process states. [5]

3. What are external and internal fragmentation. [5]

4. A single processor system has three resource types X, Y and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the table allocation denotes the number of units of each resource type allocated to each process, and the table request denotes the number of units of each resource type requested by a process in order to complete execution. Is the system in a safe state? If yes find the safe sequence. [5]

Process id	Allocation			Request		
	X	Y	Z	X	Y	Z
P1	1	2	1	1	0	3
P2	2	0	1	0	1	2
P3	2	2	1	1	2	0

5. Given memory partition of 100K,600K,200K,300K and 400 K. How would each of the First Fit, Best Fit and Worst Fit algorithm place processes of 312K ,217K, 95K and 426K . Which algorithm makes the most efficient use of memory. [5]

6. Prove Dekker-Peterson's solution ensures mutual exclusion. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Discuss about monolithic and layered approach of operating system's structure. [7]

(b) Briefly explain about the following types of operating systems: (i) Batch, (ii)Interactive, (iii) Time-sharing, (iv) Real time [8]

8. (a) List out different services of Operating Systems and explain each service. [5]

(b) What is distributed operating system? What are the advantages of distributed operating system? [5]

(c) What are system calls? Explain system calls with example? [5]

9. (a) Given references to the following pages by a program, 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults will occur if the program has three (3) page frames available to it and uses FIFO replacement strategy and LRU replacement strategy. Which replacement strategy in the above performs better and why? [10]

(b) Consider a system with a 32-bit logical address space, a two-level paging scheme, 4 byte page table entries, 1 KB pages and a 4 entry TLB. The page-table base register access time is 0 ns, TLB access time 5 ns and memory access time is 200 ns. What is the average memory access time. [5]

10. (a) Explain how semaphores achieves mutual exclusion. [5]

(b) Write down the dining Philosopher's problem. Provide its solution through monitors. [10]

11. (a) Suppose a disk drive has 300 cylinders(0 to 299). The current position of the drive is 60. Consider the track requests in the disk queue for I/O (87,170,40,150,36,72,66,15). Calculate the total head movement and average head movement for the following disk scheduling algorithms. i) SSTF scheduling ii)Scan Scheduling [10]

(b) What are the advantages and disadvantages of a two-level verses tree-structured directories. [5]

*** END OF PAPER ***